BAYORA

Securing Adversarial AI Safety Testing Infrastructure

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EXECUTIVE SUMMARY

Bayora is a zero-trust AI safety testing platform that securely isolates the **Red Team**, **Blue Team**, and **Client LLM** while enabling continuous, blind adversarial evaluations and producing tamper-evident cryptographic evidence backed by a SHA-256 Hash Chain.

1. Problem Statement & Threat Landscape

As Large Language Models (LLMs) are deployed in high-assurance enterprise environments (banking, healthcare, national infrastructure), rigorous adversarial red-teaming has become mandatory. However, existing testing setups frequently execute Red Team exploits, Blue Team defenders, and client models inside shared Docker or Kubernetes environments. This creates severe systemic vulnerabilities:

⚠️ Cross-Team Data Leakage

Red teams prematurely inspect active defense heuristics; Blue teams overfit directly on secret exploit corpora before real-world deployment.

⚠️ Model & Session Contamination

Unrestricted container sockets allow adversarial payloads to pollute multi-tenant model memory, context boundaries, or exfiltrate credentials.

⚠️ Side-Channel & Timing Attacks

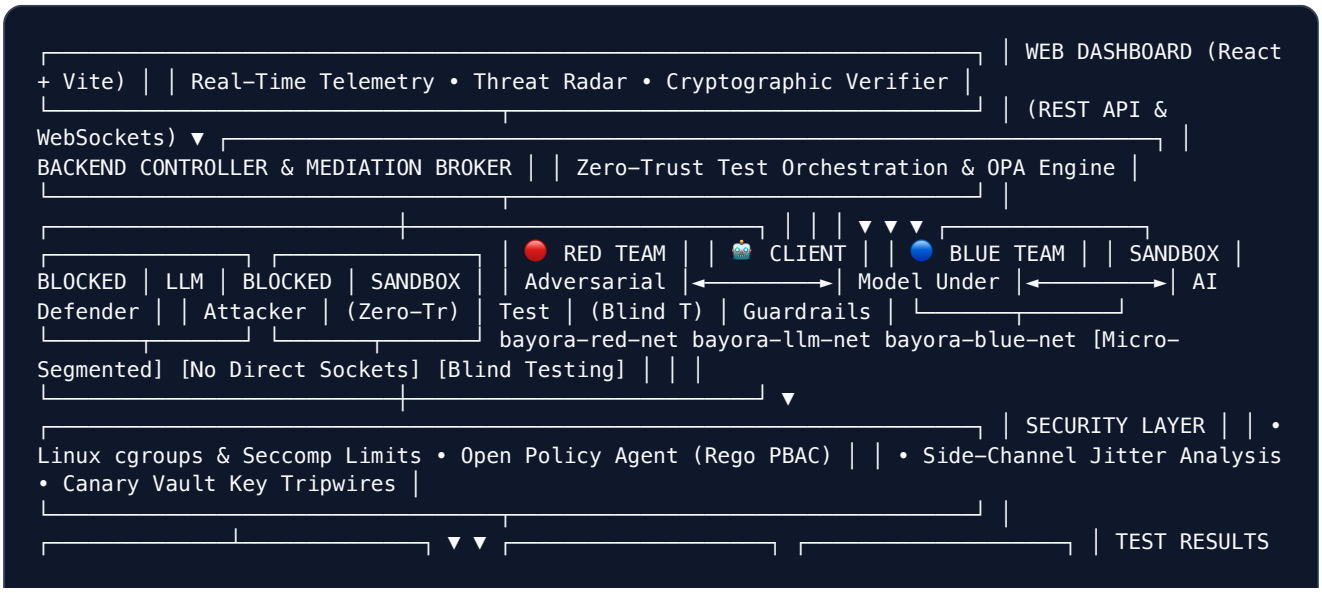
Attackers measure microsecond execution latency and CPU contention to extract confidential system prompt tokens and canary parameters.

⚠️ Unverifiable Evaluation Records

Test records and compliance logs are stored in mutable databases, making post-audit repudiation or undetected tampering a major legal risk.

2. System Architecture & Zero-Trust Isolation Flow

Bayora enforces strict network micro-segmentation and policy mediation. No workload can communicate directly with peer workloads:




TAMPER-EVIDENT
DATABASE
SHA-256 LEDGER

BAYORA

Continuous Security Loop & Verification Specification

Verification Report
27/27 Automated Tests Passing
GitHub: subwaycookiecrunch/Bayora

3. The Signature Innovation: Continuous Security Loop

Bayora automates the closed-loop safety remediation cycle: **Attack → Detect → Defend → Retest**. Discovered vulnerabilities are autonomously patched and cryptographically certified in under 15 milliseconds:

STEP 1: ATTACK
Red Sandbox
Generates jailbreak, prompt injection, or mutated Base64 obfuscated exploit payload.

STEP 2: DETECT
Baseline Probe
Unshielded evaluation reveals model compromise (e.g. secret vault canary key leaked).

STEP 3: DEFEND
Blue Auto-Patch
Blue Team engine synthesizes dynamic heuristic guardrail rule matching the exploit vector.

STEP 4: RETEST
Safety Verified
Identical exploit payload executed against updated defense; 100% remediation proven.

4. Core Pillars & Differentiators

POLICY Policy-Based Access Control (PBAC / OPA)

Open Policy Agent evaluates declarative Rego rules before packet routing. Direct Red-to-Blue peer traffic and container internet egress are strictly denied.

AUDIT Tamper-Evident SHA-256 Hash Chain

Evaluations are cryptographically linked via `prev_hash` pointers and HMAC-SHA256 signatures. Interactive verifier detects and flags database tampering down to the exact block index.

ADVERSARIAL LLM-Native Security Testing

Automated detection of DAN jailbreaks, delimiter hijacking, system prompt extraction, token smuggling, and confidential vault token exfiltration.

HARDWARE Side-Channel & Jitter Telemetry

Tracks microsecond inference duration variance to eliminate timing-based secret token extraction and noisy-neighbor cgroups CPU/memory starvation.

5. Complete Technology Stack

Layer	Technologies Used	Key Role in Bayora
Frontend	React 18, Vite, Tailwind CSS, Recharts, Lucide	Real-time dark mode cybersecurity dashboard, threat matrix, audit verifier
Backend	Python 3.11+, FastAPI, Uvicorn, SQLAlchemy	Asynchronous test orchestration, mediation controller, REST & WebSockets
Policy Engine	Open Policy Agent (OPA), Rego language	Micro-segmentation policy enforcement, zero-trust packet inspection
Security Core	SHA-256 Hash Chain, HMAC, Linux cgroups	Tamper-evident cryptographic ledger, side-channel jitter monitoring
Deployment	Docker, Docker Compose, Rootless Containers	Segmented internal networks (`bayora-red-net` , `bayora-blue-net` , `bayora-llm-net`)

Layer	Technologies Used	Key Role in Bayora
Monitoring	Prometheus Metrics (`/metrics`), Grafana-ready	Real-time CPU, RAM, active sandbox health, and isolation telemetry

6. Market Opportunity & Impact

Market Sizing: Global Generative AI Cybersecurity was estimated at **$8.9B in 2025**, reaching **$56.8B by 2033** (26.8% CAGR). India's enterprise AI security market is projected to reach **$9.92B by 2030** (44.9% CAGR).

Target Sectors: AI Startups, Enterprise Financial & Healthcare institutions handling sensitive data, Government & Defense contractors, and SaaS Providers.

Commercial Revenue Model

- **SaaS Subscription:** Tiered model evaluations per month.
- **Enterprise Licensing:** On-premises / VPC zero-trust deployment.
- **Managed AI Security:** Continuous red-team testing as a service.